

The Interpretive Graph Overlay: Modeling Case Law as Normative Events for Deterministic Legal Retrieval

Anonymous Author(s)
Submission Id: 100

Abstract

Modern legal retrieval systems reliably recover the literal wording of statutes, yet frequently fail to capture the *applicable interpretation* when courts revise their understanding of those texts over time. This paper introduces the *Interpretive Graph Overlay*, demonstrating that the SAT-Graph framework can model judicial decisions as Action entities that generate semantically distinct **Interpretive Versions** of norms, encoding their *ratio decidendi* (binding reasoning), scope, and validity intervals. By reusing SAT-Graph’s core primitives—Item, Version, Action, and TextUnit—the approach achieves ontological parsimony while enabling deterministic, time-aware retrieval. Neural agents can answer not only “What does the law say?” but also “How was it interpreted on this date?” The approach is validated through three case studies of complex normative topologies (including interpretive mutations and legislative overrides) from the Brazilian Supreme Court, with modeling patterns and structured JSON encodings provided for reproducibility.

CCS Concepts

• **Applied computing** → Law; • **Computing methodologies** → **Knowledge representation and reasoning**; • **Information systems** → *Information retrieval*.

Keywords

Legal Knowledge Graphs, GraphRAG, Case Law Modeling, Neuro-Symbolic AI, Judicial Interpretation

1 Introduction

To overcome the limitations of unstructured semantic retrieval and advance the pursuit of trustworthy Legal AI, recent research has begun to explore structured, deterministic frameworks. Notable efforts, such as the Structure-Aware Temporal Graph RAG (SAT-Graph) [6], have addressed the challenge of *Textual Determinism*—the ability of an AI system to reconstruct, with absolute precision, the wording of a legal statute as it existed on any specific historical date. These works introduced formal models that capture the form, structure, temporality, and causality of legislative texts. However, establishing the correct text is only the first step. A fundamental gap remains: the text of the law and the law as applied are not always identical.

Legal norms are targets in motion, subject to “interpretive drift.” In dynamic legal systems, a statute’s text may remain immutable while the applicable norm is radically altered. For instance, the U.S. Due Process Clause has remained textually static since 1868, yet the prevailing norm regarding abortion rights was inverted between *Roe v. Wade* and *Dobbs*. Similarly, in Civil Law jurisdictions like Brazil, courts recognize “constitutional mutations”—changes to the binding interpretation of a clause without legislative amendment.

This temporal instability of legal norms poses a fundamental challenge for automated legal reasoning systems. Standard RAG architectures, despite leveraging vast digitized corpora, remain blind to this temporal layer of validity. Even systems capable of reconstructing the literal wording of a statute at a specific date may still fail to capture the judicial interpretation that governs its actual application. To bridge this gap, this paper introduces the *Interpretive Graph Overlay*. It proposes that judicial decisions should not be modeled merely as passive documents, but as active events (represented as Action entities) within the Knowledge Graph.

This work extends the SAT-Graph framework by generalizing its core abstractions and API primitives to accommodate judicial reasoning. Just as a legislative amendment creates a new textual version of an article, a High Court decision creates an “Interpretive Version”—a conceptual entity that inherits the statute’s text but carries a distinct semantic payload (the *ratio decidendi*) and a defined validity interval. This approach leverages ontological parsimony, reusing core entities such as Item, Version, Action and TextUnit to operationalize precedents as graph events that *interpret, distinguish, or overrule* provisions. This enables Deterministic Legal Agents capable of answering not only “What does the law say?” but “How was this law interpreted on this date?”. The result is a unified Knowledge Graph where a query for a legal provision retrieves not just the legislative text, but the prevailing judicial understanding.

2 Related Work

This proposal sits at the intersection of Legal Knowledge Representation, Case-Based Reasoning (CBR), and Neural Retrieval.

Legal Ontologies and Statutory Modeling. The challenge of modeling legal norms has been extensively addressed through formal ontologies. Frameworks like LKIF-Core [9] and standard XML schemas like Akoma Ntoso [13] have provided robust methods for representing legislative structure. More recently, library reference models (IFLA LRMoo) have been integrated into legal frameworks to track the diachronic evolution of statutes, notably in SAT-Graph [6]. However, all these models—from early ontologies to recent temporal frameworks—remain “statute-centric,” treating judicial decisions as external documents linked by citation rather than constitutive events that alter the norm’s semantic substance.

Case-Based Reasoning (CBR). Beyond statutory modeling, AI & Law has a rich tradition of modeling precedents, from factor-based systems like HYPO [3] and CATO [1] to recent work on interpretive argumentation. Araszkievicz [2] conceptualizes precedents as “interpretive resources” supplying *Case Frames*. This work aligns with Araszkievicz’s theoretical stance but diverges in implementation: instead of building a specialized argumentation engine, it embeds

these interpretive structures directly into the graph’s versioning layer, making them accessible to general-purpose RAG pipelines.

GraphRAG and Validity. Retrieval-Augmented Generation (RAG) [12] grounds LLMs in external data, yet standard “flat” retrieval systems struggle with temporal reasoning [11]. Emerging GraphRAG approaches [7] leverage graph structure to improve retrieval, but lack a domain-specific model of legal validity. While frameworks like SAT-Graph address this for statutory timelines, they do not incorporate judicial decisions. This work extends that approach by introducing a deterministic mechanism to integrate precedents into the temporal graph, enabling retrieval systems to reason about both legislative changes and interpretive shifts.

3 Theoretical Foundation

The transition from modeling statutory law to case law requires a shift in perspective: from tracking *textual alterations* (amendments, repeals) to tracking *normative attribution*—how judicial decisions assign binding interpretations to legal texts. Despite this difference in subject matter, the underlying modeling structure remains the same: both processes create temporal dependencies between versions of normative content. This structural similarity enables reusing SAT-Graph’s versioning abstractions and primitives to model case law evolution.

3.1 Ontological Parsimony

Rather than introducing ad-hoc entities for judicial reasoning, this approach leverages SAT-Graph’s **Event-Centric** architecture [6], treating legislative and judicial acts as structurally equivalent: both produce temporal versions through discrete actions.

In the statutory domain, a **Temporal Version (TV)** represents the state of a norm after a legislative act (enactment or amendment). This definition extends naturally to case law: a binding precedent that changes how a statute is interpreted produces an **(Interpretable) Temporal Version (I-TV)**. The mechanism is identical in both cases. The legal document (statute or judgment) is modeled as a versioned node (*Instrument*), while the specific **micro-event** (a discrete legislative or judicial command) is formalized as an **Action** that triggers the transition from the previous version to the new temporal version. This unification allows reusing the same event topology for both legislative amendments and interpretive mutations (Figure 1).

As illustrated, the Action entity serves as the causal nexus. Its defining relationships—created (instantiating the new version) and modified (referencing the affected version)—apply equally to legislative amendments and judicial adjudications. As detailed in Table 1, the same graph structure models both domains.

Consequently, a query engine designed to traverse legislative timelines can apply the same graph traversal logic to judicial timelines. Both the precedent itself and the norms it interprets are versioned entities: subsequent rulings (e.g., motions for clarification) generate new versions of the judgment, which in turn trigger updates to the linked Interpretive Versions. Figure 2 illustrates this unified hierarchy where Statutory Versions (*S_TV*) and Interpretive Versions (*I_TV*) coexist as peers in the same derivation chain.

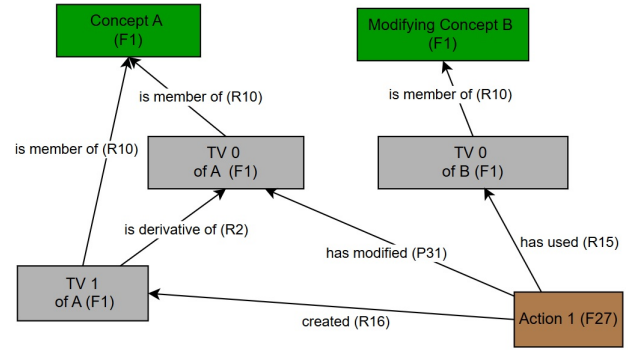


Figure 1: The Isomorphic Event Topology. The Action acts as the causal nexus between the *Instrument* (Modifying Concept B, e.g., an Amending Law (Component) or a Judgment (Component)) and the *Target* (Concept A, e.g., a Norm (Component)). The topology remains invariant: the Action uses the instrument (*TV₀* of B), acts on the target’s previous version (*TV₀* of A), and creates a new derived version (*TV₁* of A).

Table 1: Isomorphism of the Event-Centric Model

Graph Entity	Statutory Context	Interpretive Context
The Trigger (Action)	Legislative Amendment (e.g., “Law No. 11.964”)	Judicial Adjudication (e.g., “STF Judgment HC 126.292”)
The Output (Temporal Version)	A new Statutory Version (Abstract Normative State).	A new Interpretive Version (Abstract Applied Meaning).
Realization (Language Version)	Legislative Text (Natural Language Provision).	Semantic TextUnits (Ratio, Facts, Holding).

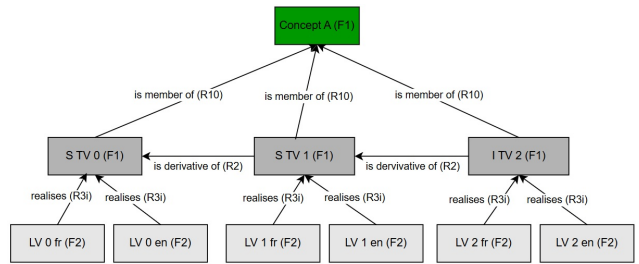


Figure 2: Unified Diachronic Chain. Statutory Versions (*S_TV*) and Interpretive Versions (*I_TV*) act as ontological peers within the same R2 is derivative of lineage. Language Versions (LV) realize each temporal version as concrete text.

3.2 Hierarchical Decomposition of Judgments

To operationalize this isomorphism, judgments must be decomposed into citable fragments, just as statutes are divided into hierarchical components (Titles, Chapters, Articles). This work extends SAT-Graph’s recursive Component-Level Modeling pattern

to judicial documents (Figure 3). The hierarchy follows the same three-tier structure: (1) **Component Concept**, the abstract identity of a structural part (e.g., “The Majority Opinion” or “The Dissenting Opinion”); (2) **Component Temporal Version**, the specific semantic content at the moment of adjudication; (3) **Component Language Version**, the concrete textual realization.

This decomposition enables semantic precision. By linking the Interpretive Temporal Version of a target provision to a specific Component Temporal Version (e.g., the majority opinion) rather than to the judgment as a whole, the model isolates the binding *ratio decidendi* from non-binding *obiter dicta*. This creates a traceable lineage between specific judicial reasoning and its normative effects on the interpreted statute.

3.3 Linking Granularity Strategies

Given that a single judgment may interpret multiple statutory provisions, the model supports two linking strategies:

Fine-Grained (Component-Level): The Action links (via R15 has used) the resulting Interpretive Version specifically to the Component Temporal Version containing the binding reasoning (e.g., the majority opinion or the convergent rationale from multiple concurring opinions). This maximizes retrieval precision by isolating the *ratio decidendi* from other parts of the judgment.

Coarse-Grained (Document-Level): As a fallback, the Action links to the judgment as a whole (the root Temporal Version). This approach is used when internal structure is not available or when the entire judgment functions as an indivisible interpretive unit.

4 The Interpretive Graph Overlay

Having established the theoretical foundation, this section describes the implementation. The Interpretive Graph Overlay builds on four canonical entities defined in the SAT-Graph API [5]:

Item Represents the abstract identity of a legal artifact (e.g., “Article 5” or “Judgment HC 123”). Items are organized hierarchically, implementing the Concept and Component Concept abstractions. **Version** Represents the temporal state of an Item at a specific point in time, implementing Temporal Version.

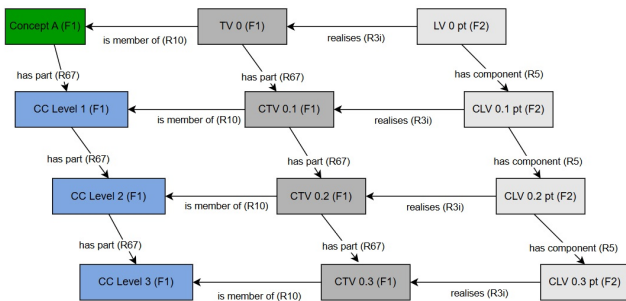


Figure 3: Component-Level Modeling. The hierarchical structure mirrors at three layers: abstract concepts (left), temporal versions (center), and textual realizations (right), enabling fine-grained linking between judgment fragments and affected norm provisions.

Action Represents the causal **micro-event** (legislative or judicial) that transitions the graph from one state to another.

TextUnit Represents a concrete text realization of a Version, implementing Language Version. It supports variations in both natural language (e.g., Portuguese, English) and semantic aspect (e.g., canonical text, summary, ratio decidendi).

This architecture allows a single Version to be associated with multiple TextUnits, each representing a different projection of the legal content. For example, a Version of a judgment may have a canonical TextUnit in Portuguese, a translated TextUnit in English, and a summary TextUnit.

4.1 Semantic Mapping via TextUnit Aspects

The challenge in modeling case law lies in decoupling logical argumentation from document structure. While a judgment’s Items reflect its organizational hierarchy (e.g., sections, opinions), the binding interpretation may emerge from specific argumentative fragments scattered across multiple structural components. The aspect property addresses this: by tagging TextUnits with semantic roles (e.g., ratio, facts, holding), the model enables direct retrieval of the *ratio decidendi* independently of its location within the judgment’s structural tree.

This allows a single structural component to be associated with multiple TextUnits, each tagged with different semantic aspects. Table 2 illustrates this mapping for a Brazilian Supreme Court judgment.

Table 2: Mapping Argumentative Roles to Aspects (Brazilian Example)

Argumentative Role	Structural Component	TextUnit Aspect
Facts	<i>Relatório</i> (Report)	facts
Ratio Decidendi	<i>Voto</i> (Vote)	ratio
Holding	<i>Dispositivo</i> (Ruling)	holding

Listing 1 illustrates the serialization of an Interpretive Version. The semantic aspects (ratio, holding) extracted from the judgment become the textual content of this interpretive version, representing how the statutory provision is interpreted after the judicial decision.

```

1 {
2   "id": "urn:lex:br:federal:constituicao:...@2016-02-17!art5_inc57_hc126292",
3   "item_id": "urn:lex:br:federal:constituicao:1988-10-05:1988@art5_inc57",
4   "type": "interpretive",
5   "validity_interval": { "start": "2016-02-17", "end": "2019-11-06" },
6
7   "metadata": { "version_type": "interpretive_mutation" },
8
9   "text_units": [
10    {
11      "aspect": "ratio",
12      "language": "en",
13      "content": "The provisional execution of a criminal conviction ..."
14    },
15    {
16      "aspect": "holding",
17      "language": "en",
18      "content": "Imprisonment is allowed after confirmation..."
19    }
20  ]
21 }
22

```

Listing 1: Example serialization of an Interpretive Version (HC 126.292)

Encapsulating Argumentation Models. A significant advantage of the `TextUnit` entity is its extensibility. While the content field stores natural language, the metadata field can store structured analytical frameworks, such as *Case Frames* [2], with slots for interpretive provisions, competing interpretations, and meta-rules. This layering allows SAT-Graph to serve as the persistence layer for advanced argumentation systems, enabling retrieval systems to access both document-level precedents and the formal reasoning structures they contain.

5 Formal Specification & API

This section demonstrates the framework’s robustness by generalizing the *SAT-Graph Primitive API* [5]. Instead of introducing separate retrieval methods for statutory and case law, this generalization reveals **Polymorphic Validity Logic**: the temporal query mechanism treats Statutory and Interpretive Versions as equivalent node types in the derivation chain.

The primitive `getValidVersions(item_id, target_date)` returns versions **active on** the target date. This reveals three patterns: (1) **Stability**: A single Statutory Version (S_{TV}) applies; (2) **Interpretive Overlay**: The S_{TV} is **modified by** an Interpretive Version (I_{TV}); (3) **Interpretive Divergence**: The S_{TV} coexists with conflicting I_{TV} s, representing splits or uncertainty.

Unified Lineage Analysis. Lineage primitives operate polymorphically across domains. Specifically, the primitive `getItemHistory(id)` retrieves a timeline that interleaves legislative amendments with judicial interpretations (e.g., *Enactment* $\rightarrow$ *Judicial Interpretation* $\rightarrow$ *Legislative Override*).

6 Case Studies: Landmark Precedents

To validate the expressiveness of the Interpretive Graph Overlay, this section analyzes complex precedents involving interpretive overlays—scenarios where the statutory text remains valid (S_{TV} is active) but its semantic scope is altered, restricted, or expanded through judicial interpretation. These cases differ from declarations of unconstitutionality with text reduction, which are topologically simpler: the judicial action invalidates the statutory version, functionally equivalent to a legislative repeal.

Case A: Interpretive Mutation (The Presumption of Innocence). *Context:* The Brazilian constitutional text regarding the presumption of innocence (Art. 5, LVII) has remained unchanged since 1988. However, the binding interpretation governing imprisonment after second-instance conviction underwent two mutations: in 2016, the Supreme Court (HC 126.292) ruled that execution of sentence did not violate the presumption; in 2019, the Court reversed itself (ADCs 43, 44, 54). *Scenario:* A user queries on June 1, 2017: “Can a defendant be imprisoned after conviction in second instance?” *Graph Traversal:* (1) The neural agent calls `getValidVersions` for 2017-06-01, which returns two active versions: the statutory version ($S_{TV_{1988}}$) and an interpretive version ($I_{TV_{2016}}$). (2) The agent retrieves the `ratio` aspect from $I_{TV_{2016}}$ and determines it represents the binding interpretation for that period. *Output:* “Yes,” citing the 2016 precedent despite the constitutional text unchanged.

Case B: Constitutional Dialogue (Legislative Backlash). *Context:* In 2016, the Supreme Court (STF) ruled Vaquejada (rodeo) unconstitutional, interpreting the prohibition of cruelty in Art. 225, §1, VII (ADI 4983). In response, Congress enacted Constitutional Amendment 96 (2017), adding §7 to Art. 225 to explicitly exempt cultural sports from the cruelty ban. *Scenario:* A user queries on January 1, 2018: “Is Vaquejada constitutional?” *Graph Traversal:* A semantic search for “Vaquejada” identifies two relevant provisions: Art. 225, §1, VII and Art. 225, §7. The agent calls `getValidVersions` for each, retrieving: (1) the original statutory version of §1, VII; (2) its interpretive version (STF: prohibition applies to Vaquejada); and (3) the new statutory version of §7 (cultural sports exemption). *Resolution:* The agent analyzes the returned versions and identifies a **Rule-Exception Topology**: the amendment functions as *lex posterior specialis*, creating an exception without invalidating the general judicial interpretation of Art. 225, §1, VII. *Output:* “Yes, provided it is registered as cultural heritage (Art. 225, §7), establishing a constitutional exception to the general prohibition in ADI 4983.”

Case C: Simultaneous Divergence (Jurisprudential Split). *Context:* Two Supreme Court Panels (Turmas) issued conflicting rulings on the constitutionality of an IPI tax credit. The First Panel (May 2018) allowed the credit, interpreting the non-cumulativity principle broadly. The Second Panel (August 2018) denied it, requiring an actual downstream tax obligation. No unifying Plenary decision exists. *Scenario:* A user queries on December 1, 2018: “Is IPI credit allowed for non-taxed outputs?” *Graph Traversal:* The neural agent calls `getValidVersions`, which returns three active versions: the original statutory version ($S_{TV_{1988}}$) and two conflicting interpretive versions ($I_{TV_{Panel1}}$ and $I_{TV_{Panel2}}$). *Resolution:* Detecting multiple active interpretive versions overlaying the statute, the agent identifies a **jurisdictional split**. The statutory text remains valid but lacks a unified binding interpretation. *Output:* “There is no consensus. The First Panel allows the credit (RE 607.109), while the Second Panel denies it (RE 566.819). This divergence creates legal uncertainty pending Plenary resolution.”

7 Generalization across Jurisdictions

Although this work uses Brazilian law as the primary case study, the Interpretive Graph Overlay is designed for cross-jurisdictional applicability. Despite terminological differences, the underlying structure remains constant: courts interpret statutes, interpretations gain binding force, and legislatures respond through amendments. Table 3 demonstrates how the abstract graph entities map to diverse legal systems.

Adapting to Common Law (USA/UK/India). In common law areas without statutory basis (e.g., Torts), precedents create their own normative targets. To model cases like *Donoghue v. Stevenson*,

Table 3: Cross-Jurisdictional Mapping of Graph Entities

Graph Entity	Common Law (USA, UK, India)	Civil Law (Europe)	Guiding Cases (China)
Instrument	Court Opinion	Décision / Urteil	Guiding Case
Target	Statute / Precedent	Code Article	Statute
Binding Part	Ratio Decidendi	Considérannts Gründe	/ Main Points

the graph creates abstract Items representing judge-made legal principles (e.g., “duty of care”), deriving their content from the opinion’s *ratio decidendi*. Furthermore, *distinguishing* is modeled not as invalidation but as scope refinement: the Action creates a new interpretive version with narrowed applicability conditions, while the original precedent remains valid for cases meeting the broader criteria.

Centralized Review (France). In centralized review systems, constitutional courts often preserve a statute’s validity conditional upon a specific reading, rather than declaring outright invalidation. The French *réserve d’interprétation* (interpretive reservation) exemplifies this technique [15], used in both *a priori* review and the *a posteriori* QPC procedure introduced in 2008. The model represents this precisely: the court decision functions as an Action creating an Interpretive Version that overlays the statute, binding future application to the court’s specified reading (extracted from the *considérants*).

Structured Guidance Systems (China). Chinese *Guiding Cases* follow a standardized structure [8, 10] where “Main Points” sections serve as pre-extracted *ratio decidendi*. Since these cases have persuasive rather than binding authority, the model can use Action metadata (e.g., *authority_level*: “persuasive”) to distinguish them from mandatory precedents. This metadata is returned alongside the interpretive version, enabling neural agents to reason about the appropriate weight to assign to the guidance.

Handling Federalism (Circuit Splits). In federal systems, precedent validity varies by jurisdiction. To model circuit splits [4], the Action entity can include *geographic_scope* metadata (e.g., “9th Circuit”). When *getValidVersions* returns multiple conflicting interpretive versions for the same statute, the metadata enables neural agents to identify which interpretation applies in the target jurisdiction and recognize conflicting authority elsewhere.

8 Conclusion and Future Work

This work introduced the **Interpretive Graph Overlay**, extending the SAT-Graph framework to model case law using the same abstractions and primitives developed for statutory versioning. By treating precedents as versioned entities that generate (Interpretive) Temporal Versions, the approach enables RAG systems to distinguish between statutory text and its authoritative judicial interpretation. This demonstrates ontological parsimony: rather than introducing domain-specific entities for case law, the framework reuses graph structures by generalizing their semantic scope.

This work also contributes to the explainability of AI-assisted legal reasoning. By explicitly modeling interpretive provenance—linking each normative conclusion to the specific judicial decision that established it—the framework enables retrieval systems to provide traceable citations. This addresses a key limitation of purely neural approaches, which struggle to ground interpretive claims in authoritative sources.

8.1 Addressing Limitations

A deterministic graph is necessary, but not sufficient, for reliable legal reasoning. While the overlay models explicit overruling, it does not capture *implicit overruling*, where a precedent becomes

obsolete without formal revocation. The framework assumes determinism requires explicitness: implicit overruling must be detected at the ingestion layer through expert curation or citation network analysis, rather than encoded as heuristic inference rules in the graph. This design prioritizes precision and auditability—ensuring the graph only represents formally established relationships—over maximizing coverage of potentially obsolete precedents.

Future Work. While this work demonstrates the applicability of SAT-Graph abstractions and primitives to case law through modeling patterns and case studies, comprehensive quantitative evaluation against established legal retrieval benchmarks (e.g., LegalBench-RAG [14]) represents the next step. Additionally, scaling the approach requires developing semi-automated ingestion pipelines that leverage LLMs to extract interpretive structures from broader jurisdictional corpora.

Data and Artifact Availability

To support reproducibility, the structured JSON encodings of the three case studies (HC 126.292, ADI 4983, and IPI Divergence) are available in the anonymous repository: <https://anonymous.4open.science/r/sat-graph-caselaw-0F74/>. These artifacts illustrate the application of the framework described in Section 6.

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